

Student's Name: _____ Teacher's Code : _____



Saint Ignatius' College, Riverview

Mathematics Assessment Task

2021

Year 12
Mathematics
Task 4
Trial HSC Replacement Task
Date : 27 th August 2021

General Instructions:		Format:	
• Reading time : 5 minutes • Time Allowed: 2 hours • Write using blue or black pen only • NESA approved calculators may be used • Attempt all questions in the space provided on the paper • Write your name and your teacher's code in the positions indicated • Marks may not be awarded for missing or carelessly arranged working.			
Teachers :			
• Mr N Mushan	NHM	Section A 14 Marks	
• Mr P Collins	PPC	Short Answer	
• Dr M Furtado	MXF	Section B 15 Marks	
• Mrs F Yates	FEY	Short Answer	
• Mr J Newey	JPN	Section C 16 Marks	
		Short Answer	
		Section D 11 Marks	
		Short Answer	
		Section E 14 Marks	
		Short Answer	
		Total	70 Marks

SECTION A (14 Marks)**(Start a new booklet)**

(a) Find the equation of the tangent to the curve $y=(1-5x)^{-1}$ at the point (0, 1). **(3)**

(b) Differentiate $y=\frac{\ln(x^3)}{2}$ with respect to x . **(2)**

(c) Differentiate $y=\frac{e^{2x}}{\cos x}$ with respect to x . **(2)**

(d) Evaluate $\int_0^{\frac{\pi}{3}} (x + \sin x) dx$. **(2)**

(e) The function $y = f(x)$ is continuous for all values of x . **(2)**

The following is known of the function's properties.

- $x > -2$, $f'(x) < 0$, $f''(x) < 0$
- $x < -2$, $f'(x) < 0$, $f''(x) > 0$
- $f(-2) = f'(-2) = f''(-2) = 0$

Sketch the graph of $y = f(x)$, labelling any x -intercepts **in your own booklet**.

(f) Solve the equation $\ln x - \frac{6}{\ln x} = 1$ **(3)**

END OF SECTION A

SECTION B (15 Marks)**(Start a new booklet)**

- (a) An infinite geometric series has a first term of 12 and a limiting sum of 15. (2)

What is the common ratio?

- (b) A function $f(x)$ is given by:

$$f(x) = \begin{cases} \frac{3}{4}(x-2)(4-x) & \text{if } 2 \leq x \leq 4 \\ 0 & \text{if } x < 2 \text{ or } x > 4 \end{cases}$$

- (i) Show this curve is a probability density function. (2)

- (ii) Find the Mode. (2)

- (c) (i) Draw the graph of $y = -\sqrt{4-x^2}$ and shade the area bounded by the function and the x axis where $x \leq 0$. (2)

The table gives the values of $y = -\sqrt{4-x^2}$ for x .

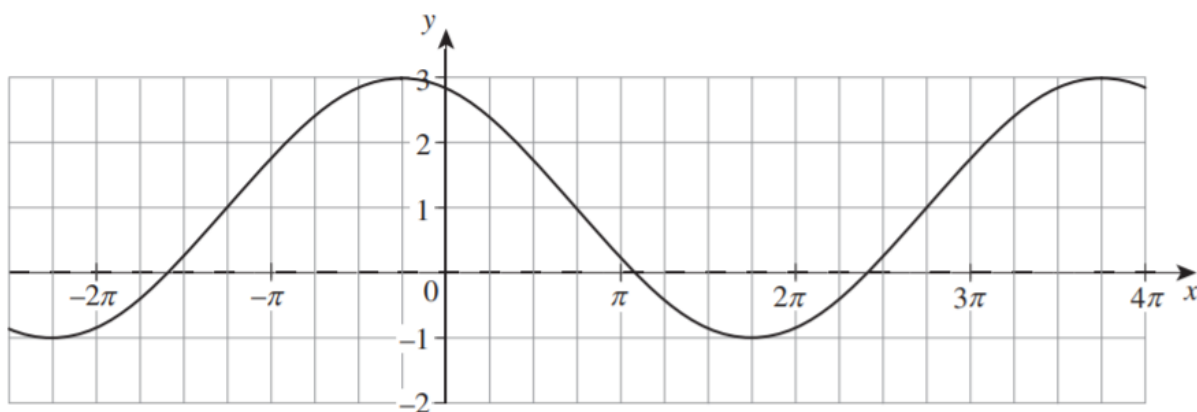
x	-2	-1.5	-1	-0.5	0
y	0	-1.3	-1.7	-1.9	-2

- (ii) By using the above table of values or otherwise, use the trapezoidal rule with 4 sub-intervals to find an approximation of the shaded area. (2)

- (iii) How could a better approximation of the shaded area be obtained using the trapezoidal rule? (1)

SECTION B is continued on next page

- (d) The graph is a function of the form $y = k \sin(a(x+b)) + c$.



Consider the equation of the function, to determine the values of k , a , b and c (4)

END OF SECTION B

SECTION C (16 Marks) (Start a new booklet)

- (a) Let X be a normally distributed random variable with mean 6 and variance 9 and let Z be the random variable with the standard normal distribution.

(i) Find $P(X > 6)$. (1)

(ii) Find b such that $P(X > 8) = P(Z < b)$. (2)

- (b) For events A and B from a sample space, $P(A|B) = \frac{1}{5}$ and $P(B|A) = \frac{1}{4}$.
Let $P(A \cap B) = p$.

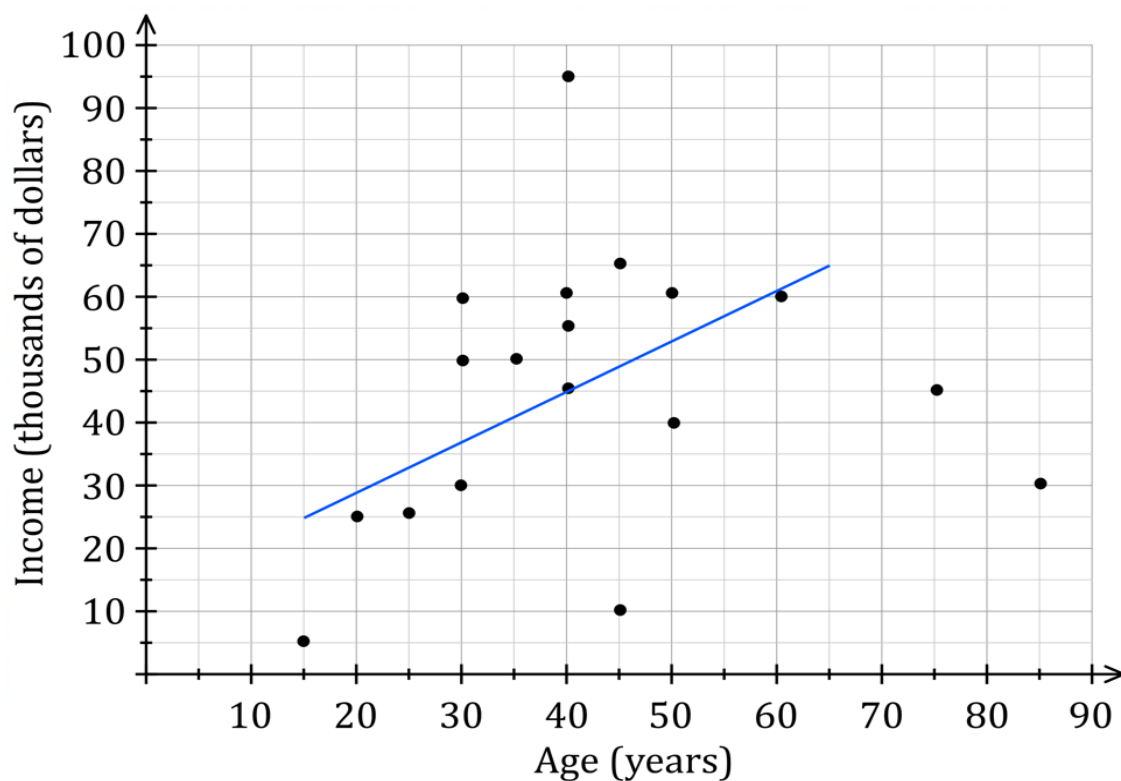
(i) Find $P(A)$ in terms of p . (1)

(ii) Find $P(A' \cap B')$ in terms of p . (2)

(iii) Given that $P(A \cup B) \leq \frac{1}{5}$, state the largest possible interval for p . (2)

SECTION C is continued on next page

- (c) A sample of 18 randomly selected people were asked to state their Age (in years) and their income (in \$1000). The results are displayed in the scatterplot below with the line of best fit drawn through (15, 25) and (65, 65).



- (i) Calculate the gradient of the line of best fit shown. (2)
- (ii) What is the equation of the line of best fit shown? (1)
- (iii) Use your equation to calculate the expected income of an 80-year-old person. (1)
- (iv) Comment on your result in part (iii). (1)
- (d) (i) Without using calculus, sketch $y = \ln x$ in your booklet. (1)
- (ii) On the same sketch in part (a), find, graphically, the number of solutions to the equation $\ln x - x = -2$. State the number of solutions in your booklet. (2)

END OF SECTION C

SECTION D**(11 Marks)****(Start a new booklet)**

- (a) A chemical substance being made in a laboratory will be airlifted by helicopter to a hospital immediately after it is completed. The substance decomposes and the amount M in kilograms present at any time t hours is given by $M = Ae^{kt}$ where A and k are constants.
- If $\frac{1}{4}$ of the mass of this substance is present after 4 hours and 4 kg of this substance will reach the hospital in 6 hours.

(i) Show the exact value of $k = -\frac{1}{4}\ln(4)$ (2)

(ii) Find the value of A , the original mass of the chemical substance. (2)

- (b) An open cylindrical water tank has base radius x metres and height h metres.

Each square metre of the base costs a dollars to manufacture and each square metre of the curved surface costs b dollars, where a and b are constants.

The combined cost of the base and curved surface is c dollars.

(i) Find c in terms of a , b , x and h . (1)
(Note that the curved surface has area $2\pi xh$.)

(ii) Show that the volume V of the tank in cubic metres is given by (2)

$$V = \frac{x}{2b}(c - \pi ax^2)$$

(iii) If x can vary, prove that V is maximised when the **cost of the base** is $\frac{c}{3}$ dollars. (4)

END OF SECTION D

SECTION E**(14 Marks)****(Start a new booklet)**

- (a) Alison decides to start saving for a deposit for a new apartment by investing \$1000 into a bank account at the start of every month. Interest is paid at a rate of 3% per annum compounded monthly.

- (i) Show that the amount in A_n in the account after n months is: (2)

$$A_n = 401000(1.0025^n - 1)$$

- (ii) Calculate the amount in the account at the end of 12 months. (1)
Give your answer to the nearest dollar.

- (iii) Determine how many deposits in total Alison will have made when her account first exceeds \$60 000. (2)

- (iv) After twelve months Alison changes her contributions to \$1500 per month. (2)
Determine if Alison will have saved \$60 000 by the end of the third year.
Justify your answer.

- (b) (i) Show that $x = \frac{\pi}{3}$ is a solution of $\sin x = \frac{1}{2} \tan x$ (1)

- (ii) On the same set of axes, sketch the graphs of the functions (2)

$$y = \sin x \text{ and } y = \frac{1}{2} \tan x \quad \text{for } -\pi \leq x \leq \pi.$$

- (iii) Hence find all solutions of $\sin x = \frac{1}{2} \tan x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$. (2)

- (iv) Use your graphs to solve $\sin x \leq \frac{1}{2} \tan x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$. (2)

END OF PAPER

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SUGGESTED SOLUTIONS

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		Short Answer	
		Total	70 Marks

SECTION A

Worked Solution

Marker's Comments

$$\textcircled{a} \quad y = (1-5x)^{-1}$$

$$y' = -1(1-5x)^{-2} \times -5$$

$$= 5(1-5x)^{-2} \quad \text{or} \quad \frac{5}{(1-5x)^2} \quad \checkmark$$

When $x=0$

$$m_{\text{tangent}} = \frac{5}{(1-5 \times 0)^2} = 5 \quad \checkmark$$

equation of tangent

$$y-1 = 5(x-0)$$

$$y-1 = 5x$$

$$\boxed{y = 5x + 1} \quad \text{or} \quad 5x - y + 1 = 0 \quad \checkmark$$

1 mark for differentiating

1 mark for gradient of tangent

1 mark for equation of tangent.

$$\textcircled{b} \quad y = \frac{\ln(x^3)}{2}$$

$$= \frac{1}{2} \ln(x^3)$$

$$y' = \frac{1}{2} \times \frac{3x^2}{x^3}$$

$$= \frac{3x^2}{2x^3}$$

$$= \boxed{\frac{3}{2x}} \quad \checkmark$$

reasonably well done

Alternative Solution

$$y = \frac{3 \ln x}{2} \quad \checkmark$$

$$y' = \frac{3}{2} \times \frac{1}{x}$$

$$y' = \frac{3}{2x} \quad \checkmark$$

many failed to simplify expression.
 $\therefore$ lost a mark.

$$\textcircled{c} \quad y = \frac{e^{2x}}{\cos x}$$

let $u = e^{2x}$

$v = \cos x$

$u' = 2e^{2x}$

$v' = -\sin x \quad \checkmark$

$$y' = \frac{2e^{2x} \cos x + e^{2x} \sin x}{\cos^2 x} \quad \checkmark$$

1 mark for correctly finding u' and v' .

correctly substituting into product rule.

$$y' = \boxed{\frac{e^{2x} (2 \cos x + \sin x)}{\cos^2 x}}$$

- some forgot v^2

- some mixed up signs.

Section A (continued)

Marker's Comments

d) $\int_0^{\frac{\pi}{3}} (x + \sin x) dx$

$$= \left[\frac{x^2}{2} - \cos x \right]_0^{\frac{\pi}{3}}$$

$$= \left(\frac{(\frac{\pi}{3})^2}{2} - \cos \frac{\pi}{3} \right) - \left(\frac{0^2}{2} - \cos 0 \right)$$

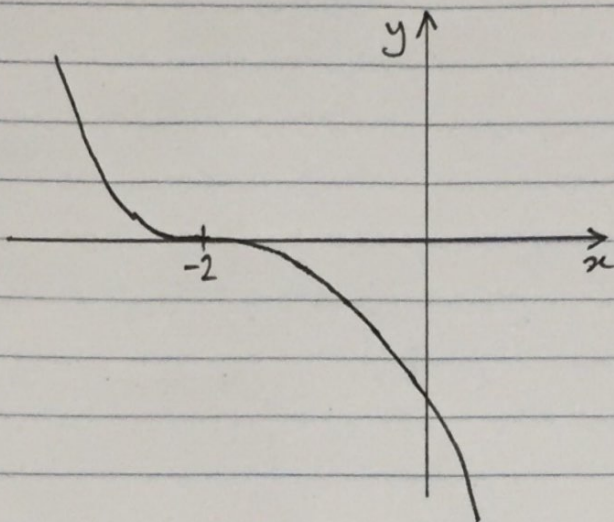
$$= \frac{\pi^2}{18} - \frac{1}{2} + 1$$

$$= \boxed{\frac{\pi^2}{18} + \frac{1}{2}} \quad \text{or} \quad \frac{\pi^2 + 9}{18}$$

✓ integrating

many messed up the arithmetic.

e)



- had to show that there is a stationary point at $x = -2$ (-30)

- concave up to left of $x = -2$

- concave down to right of $x = -2$

✓ correct shape

✓ clearly showing horizontal point of inflection at $x = -2$ ie it must be obvious.

f) $\ln x - \frac{6}{\ln x} = 1$

$$(\ln x)^2 - 6 = \ln x$$

$$(\ln x)^2 - \ln x - 6 = 0$$

$$(\ln x - 3)(\ln x + 2) = 0$$

$$\ln x = 3 \text{ or } -2$$

$$\text{ie } \log_e x = 3 \text{ or } \log_e x = -2$$

$$\therefore x = e^3 \text{ or } x = e^{-2}$$

✓ 1 mark for multiplying through by x and forming correct quadratic. (note $(\ln x)^2 \neq \ln(x^2)$)

Solving quadratic

two correct solutions.

Section B

Solutions & Comments

a) $a = 12$
 $S_{\infty} = 15$

$$S_{\infty} = \frac{a}{1-r}$$

$$15 = \frac{12}{1-r}$$

$$15 - 15r = 12$$

$$-15r = -3$$

$$r = \frac{1}{5}$$

* Well done by all

b) Proving a PDF

1) Prove $\int_2^4 f(x) dx = 1$

$$\int_2^4 \frac{3}{4} (x-2)(4-x) dx$$

$$= \frac{3}{4} \int_2^4 -x^2 + 6x - 8 dx$$

$$= \frac{3}{4} \left[-\frac{x^3}{3} + 3x^2 - 8x \right]_2^4$$

$$= \frac{3}{4} \left[\left(-\frac{64}{3} + 48 - 32 \right) - \left(-\frac{8}{3} + 12 - 16 \right) \right]$$

$$= \frac{3}{4} \left[-\frac{16}{3} + \frac{20}{3} \right]$$

$$= \frac{3}{4} \times \frac{4}{3}$$

$$= 1$$

∴ PDF

AND

need to show $f(x) \geq 0$

NO ONE DID!

$$f(x) = f(4) = 0$$

$$f(x) > 0 \quad 2 < x < 4$$

$$f(x) \geq 0 \quad 2 \leq x \leq 4$$

2 marks awarded for showing this

All to note this BY the HSC

11) Mode

- highest pt on $f(x)$

in domain

method one

$$f(x) = \frac{3}{4} (x-2)(4-x)$$

$$= \frac{3}{4} (4x - x^2 - 8 + 2x)$$

$$= \frac{3}{4} (-x^2 + 6x - 8)$$

$$\text{Axis of Sym} = -\frac{b}{2a}$$

$$= -\frac{6}{-2}$$

$$x = 3$$

concave

Down

parabola!!

∴ Mode is 3.

OR use calculus

method Two

$$f(x) = \frac{3}{4} (-x^2 + 6x - 8)$$

$$f'(x) = \frac{3}{4} (-2x + 6)$$

$$\frac{3}{4} (-2x + 6) = 0$$

$$-2x = -6$$

$$x = 3$$

$$f''(x) = -\frac{3}{2} < 0 \quad \text{max.}$$

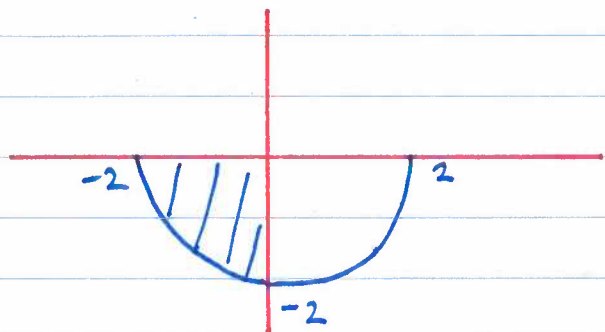
* not well done

* many found the y value not x value

* some confused with 'median' process

c) i) $y = -\sqrt{4-x^2}$

semi circle !!



ii) Trapezoidal Rule

$$A = \frac{h}{2} [f(a) + 2(f(\dots)) + f(b)]$$

$$= \frac{0.5}{2} [0 + 2(1.3 + 1.7 + 1.9) + 2]$$

$$\hat{=} 2.95 \text{ units}^2$$

iii) More Sub intervals or function values req'd

- this is how to make trapezoidal rule more effective!

Learn Your Basic Graphs!!

* Atrocious!!

* ✓ semicircle

* ✓ shaded

LHS

- failure by many to see it was a semi circle!!

* too many weren't sure

how to deal with

negatives!

IGNORE them!!

* done ok by most!

* Not too badly done!!

d) a few diff options!!

Most Popular

$$a = \frac{1}{2}$$

$$b = \frac{5\pi}{4}$$

$$K = 2$$

$$c = 1$$

Also

$$a = -\frac{1}{2}$$

$$b = -\frac{3\pi}{4}$$

$$K = 2$$

$$c = 1$$

Also

$$a = \frac{1}{2}$$

$$b = -\frac{3\pi}{4}$$

$$K = -2$$

$$c = 1$$

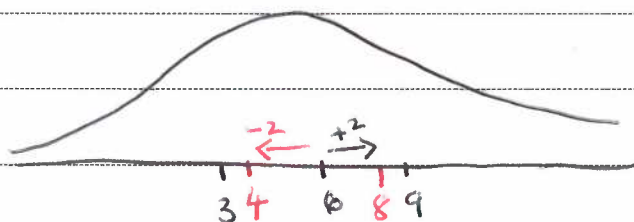
SECTION C

16 MARKS.

Marker's

(a) (i) $P(X > 6)$ $\mu = 6, \sigma^2 = 9 \therefore \sigma = 3$

Comments.



(i) $P(X)$ $P(X > 6) = 50\%$ or 0.5 .

✓ Its 50-50 from the median

(ii) $P(X > 8) = P(Z < b) = P(X < 4)$ by symmetry

from above: $x = 4$

✓ Not many

$$\therefore Z = \frac{x - \mu}{\sigma} = \frac{4 - 6}{3} = -\frac{2}{3}$$

realised that

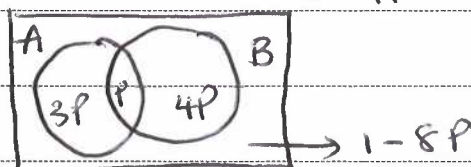
$$\therefore b = -\frac{2}{3}$$

✓ you had to find Z

(b) (i) $P(A) = ?$ $P(B|A) = \frac{P(A \cap B)}{P(A)}$

$$\therefore P(A) = \frac{P(A \cap B)}{P(B|A)} = \frac{P}{\frac{1}{4}}$$

(ii) $P(A' \cap B')$



$$= 4P$$

✓

$P(A) = 4P \therefore P(B) = 5P$

$$\therefore P(A' \cap B') = 1 - 8P$$

OR $P(A' \cap B') = P(A \cup B)' = 1 - P(A \cup B)$

$$= 1 - 8P.$$

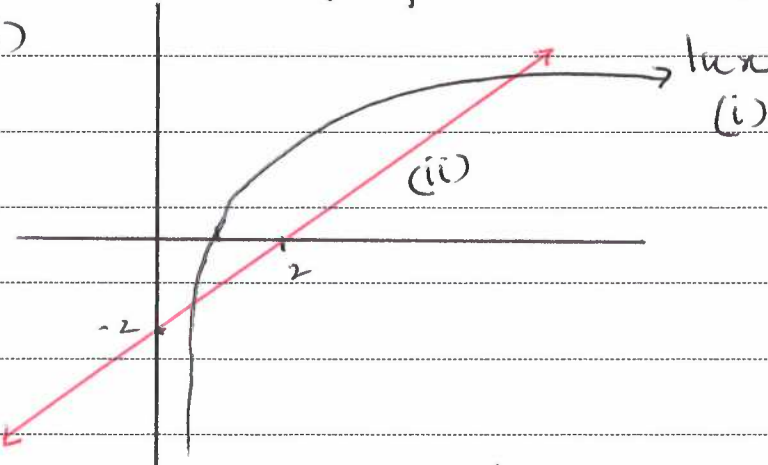
✓ Not well done!

(iii) $0 \leq 8P \leq \frac{1}{5} \quad P \neq 0$

$$\therefore 0 < P \leq \frac{1}{40} \quad \text{OR } P = (0, \frac{1}{40}]$$

✓

You may ask for an extra Writing Booklet if you need more space to answer this question.

Section C (cont.)	16 MARKS	Marker's
(C) (i) Accepted Gradient as 0.8 or 0.75	Comments.	Overall this
where appropriate working was shown	question was well answered.	
$\left. \begin{matrix} (65, 65) \\ (15, 25) \end{matrix} \right\} m = \frac{65-25}{65-12} = 0.8$	✓ ✓	
OR $\left. \begin{matrix} (60, 60) \\ (40, 45) \end{matrix} \right\} m = \frac{60-45}{60-40} = 0.75$		
(ii) $y = mx + b$ using appropriate points	✓	
$y = 0.8x + 13$	OR $y = 0.75x + 15$	
(iii) $y = 0.8 \times 80 + 13$	$y = 0.75 \times 80 + 15$	✓
$y = 77k$	$y = 75k$	
$= \$77000$	$= \$75000$	
(iv) With age Salary increase is not true for an 80 year old. or	✓	
Cannot extrapolate this equation.		
(d)	Sketch the lux graph	
	Solutions had to be seen on the graph for 2 marks.	
Soln: 2 two		
* Just stating 2 solutions without a graph no marks.		

You may ask for an extra Writing Booklet if you need more space to answer this question.

2021 ZU ADV FINAL TASK
SECTION D

Comments

(a) (i) $M = Ae^{Kt}$
 $\frac{A}{4} = Ae^{Kt}$
 $\frac{1}{4} = e^{Kt}$ ✓

$M = \frac{1}{4}A, t = 4$

$\ln\left(\frac{1}{4}\right) = 4K$

$\ln(4^{-1}) = 4K$ ✓

$-\ln(4) = 4K$

$\therefore K = -\frac{1}{4}\ln(4)$

(ii) $M = Ae^{-\frac{1}{4}\ln(4)t}$
 $4 = Ae^{-\frac{6}{4}\ln(4)}$ ✓

$M = 4, t = 6$

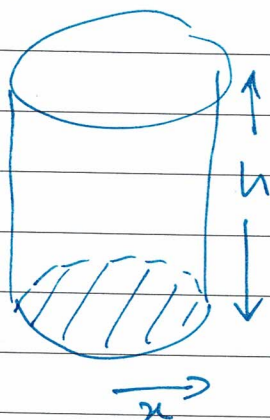
$\therefore A = \frac{4}{e^{-\frac{3}{2}\ln(4)}}$

$= 32$ ✓

Can't use Physics formulae "show that" in Part (i) means the examiner must be able to follow every step.

- Many poor attempts at very straight-forward questions
 - Use the value for K given in (i) for (ii)

(b) (i)



$\text{COST (BASE)} = a \times \pi x^2$

$\text{COST (CURVE)} = b \times 2\pi xh$

It is an open cylinder.

"each square metre of the base costs a"

(Cost Total) $C = a\pi x^2 + 2b\pi xh$ ✓

(ii) $Vol = \pi x^2 h$ $\left[\rightarrow \text{No } h \text{ in given answer} \right]$
 $V = x \times \pi xh$

It is not a cone

$\therefore 2b\pi xh = C - a\pi x^2$
 $\pi xh = \frac{C - a\pi x^2}{2b}$ ✓

OR $h = \frac{C - a\pi x^2}{2b\pi x}$

Please turn over

D(bxii) continued.

$$\therefore V = \frac{x(C - a\pi x^2)}{2b} \quad \checkmark$$

$$V = \frac{x}{2b} (C - a\pi x^2)$$

iii $\therefore V = \frac{1}{2b} (Cx - a\pi x^3)$

$$\therefore V' = \frac{1}{2b} (C - 3a\pi x^2) \quad \checkmark$$

for MAX/MIN let $V' = 0$

$$\therefore \frac{1}{2b} (C - 3a\pi x^2) = 0$$

$$C - 3a\pi x^2 = 0 \quad \checkmark$$

$$-3a\pi x^2 = -C$$

$$a\pi x^2 = \frac{C}{3} \quad \checkmark$$

$\therefore$ Cost of base = $\frac{C}{3}$ as required

$$\text{Now, } V'' = \frac{1}{2b} (-6a\pi x)$$

$$< 0 \text{ as } a, b, x > 0 \quad \checkmark$$

$\therefore$ Max value when

$$a\pi x^2 = \frac{C}{3}$$

Substitution:

This is by far the easiest method - Product rule next
*It is not quotient rule as $2b$ is a constant term w.r.t. x .

Could also use 1st derivative test but values are required (tough arithmetic/algebra). Reason for statement $V'' < 0$ required (not just a coin toss!).

SECTION E.

a) i) After month.....

$$A_1 = 1000(1.0025)$$

$$A_2 = (A_1 + 1000)(1.0025)$$

$$= 1000(1.0025)^2 + 1000(1.0025) \leftarrow \text{Ink}$$

$$\therefore A^n = 1000(1.0025)^n + \dots + 1000(1.0025)$$

$$= 1000[1.0025^n + 1.0025^{n-1} + \dots + 1.0025]$$

$$= 1000[1.0025 + 1.0025^2 + \dots + 1.0025^n]$$

$$= 1000 \left[\frac{1.0025(1.0025^n - 1)}{1.0025 - 1} \right] \leftarrow \text{Ink}$$

$$= 401000(1.0025^n - 1)$$

Well done

ii) let $n = 12$

$$A_{12} = 401000(1.0025^{12} - 1)$$

$$= \$12197 \leftarrow \text{Ink}$$

Well done

$$\text{iii) } 60000 = 401000(1.0025^n - 1)$$

$$0.1496 = 1.0025^n - 1$$

$$1.0025^n = 1.1496$$

$$n = 55.8$$

$$\therefore 56 \text{ DEPOSITS} \leftarrow \text{Ink}$$

Well done

Working required

$$N) \text{ TOTAL} = (A_1) + (A_2)$$

$$A_1 = 12197 (1.0025)^{24}$$

$$= \$12950.25 \quad \leftarrow \text{lmk}$$

$$A_2 = \frac{1500(1.0025(1.0025^{24} - 1))}{1.0025 - 1}$$

$$= \$37146.86 \quad \leftarrow \text{lmk}$$

$$\therefore \text{TOTAL} = \$50097$$

$\therefore$ No!!

Poorly done

Too many ignored (A_1) .
Assumed it earned No Interest

$$b) i) \text{ When } x = \frac{\pi}{3}$$

$$\sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

$$\frac{1}{2} \tan \frac{\pi}{3} = \frac{1}{2}(\sqrt{3})$$

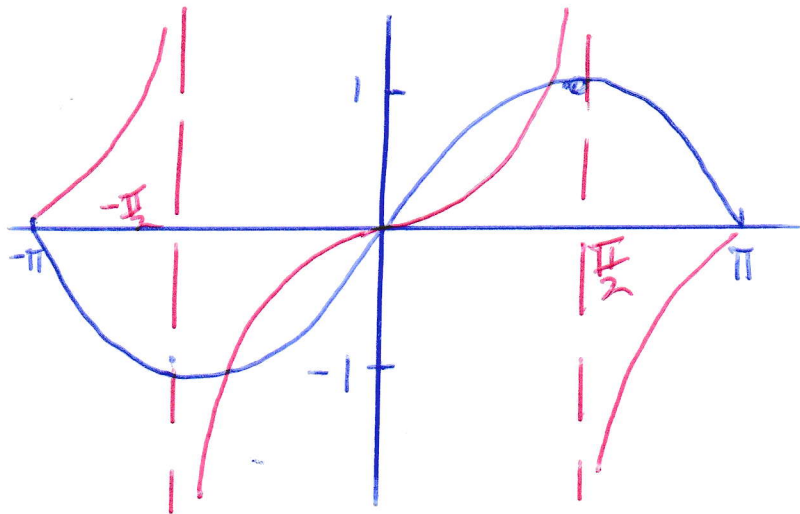
$$= \frac{\sqrt{3}}{2}$$

} $\leftarrow$ lmk

$$\therefore \text{ at } x = \frac{\pi}{3} \quad \sin x = \frac{1}{2} \tan x$$

Many went to a lot of trouble to do something so simple

ii)



$$y = \sin x$$

$$y = \frac{1}{2} \tan x$$

Well done
1mk for
each Fully
correct
graph.

• Some did
not use
correct domain

iii) $\sin x = \frac{1}{2} \tan x$

$$x = -\frac{\pi}{3}, 0, \frac{\pi}{3} \leftarrow$$

Too many
missed $x=0$
3 solns = 2mk
2 solns = 1mk

iv) $\sin x \leq \frac{1}{2} \tan x$ for
 $-\frac{\pi}{2} < x < \frac{\pi}{2}$

Poorly
done

$$-\frac{\pi}{3} \leq x \leq 0 \leftarrow \text{1mk}$$

$$\frac{\pi}{3} \leq x < \frac{\pi}{2} \leftarrow \text{1mk}$$